Heterogenous Patterns in Black Infant Health Following the Police Killing of Amadou Diallo

Extended abstract submitted for consideration at the PAA Annual Meeting 2027

AP Pittman[1], Michael Cao[1], Daichi Hibi[2], Fatima Fairfax[1], Simone Wien[3], Isabella Bouklas[1], Ralph Catalano[4], Tim A. Bruckner[3], Hedwig Lee[1]

[1] *Department of Sociology, Duke University, Durham, NC 27708*
[2] *Department of Sociology, Washington University in St. Louis, St. Louis, MO 63130*
[3] *Department of Health, Society, and Behavior, University of California, Irvine, Irvine, CA 92697*
[4] *School of Public Health, University of California, Berkeley, Berkeley, CA 94720*

Abstract:
The American policing environment is an understudied determinant of Black reproductive and perinatal health, with consequences that may vary by immigrant status. Although police violence is salient to both native-born and immigrant Black communities, its health effects may differ because these populations are embedded in distinct social and structural contexts. We examine changes in the birth sex ratio (i.e., M:F) following the killing of Amadou Diallo (a Black immigrant), using a decline in male relative to female births as an indirect indicator of excess male fetal loss. We find a 17% decline in the expected number of male births to foreign-born non-Hispanic Black mothers during the two months following the killing. In contrast, no comparable change was detected among native-born non-Hispanic Black mothers. These findings suggest that exposure to police violence may have reproductive consequences that vary across heterogeneous social contexts within the Black population and specific historical contexts.

INTRODUCTION

Historic gaps in the survival chances of Black versus White infants remain despite decades of progress, a reminder of the enduring nature of Black-White racial disparities in infant health in the United States (1, 2). While these racial disparities have remained relatively fixed through time, individual-level explanations for these gaps remain elusive (3). This circumstance has led to the identification of socioecological antecedents of infant health inequities (4). Among them is the American policing environment, which has been theorized as a manifestation of structural racism (5, 6). Black populations experience not only elevated police presence in their communities (7), but also racially disproportionate risks of stops, arrests, and fatal aggression (8, 9) that in turn affect perceptions of trust and safety (10). Exposure to police violence, therefore, appears plausible as a measure and social determinant of racial disparities in infant health (10–12).

Generally, a model of policing as an ecological exposure suggests that incidences of police violence may propagate an undue "surveillance stress" among parents (13), which may then cause *in utero* changes. Parent-level exposure to police violence, either directly or indirectly via social contagion, produces negative emotional and physiological reactions, as well as disruptions in social capital (10, 14, 15). When faced with highly public incidents of police violence that demonstrate that even the unarmed can be faced with mortal threat, Black parents in particular may worry about the life chances of their families and children (9). This, on top of a high preexisting allostatic load associated with parental and minority status (16, 17), means that pregnant Black parents responding adversely to police violence may experience acute psychosocial stress known to affect fetal development (18).

Empirical work supports the hypothesis that parental exposure to police-related stress has an adverse effect on infant outcomes. Residence by birthing parents in neighborhoods with high police exposure is associated with an 83% increase in odds of preterm birth (PTB) relative to those in low exposure neighborhoods (19). In addition, gestational exposure to fatal police violence at the population-level is associated with greater increases in PTB (20) and lost pregnancies to Black parents relative to White parents (21).

However, existing research underexplores the heterogeneity among Black parents, even though the racialized effects of police violence on infant health may differ based on within-group characteristics. Existing work on health disparities in general tends to view the Black population as a monolith (22). While the stress-pathway model, described above, appears viable in a general sense, not enough is known about how Black parents, depending on their backgrounds, may vary in their negotiation of police violence. This kind of disaggregation is important, because it enables us to understand how specific social identities may complicate the model of police violence as an ecological determinant of health. Here, we focus on immigration status as one particular kind of heterogeneity in Black populations, examining its effect on responses to a specific and highly visible type of police violence on a Black immigrant.

This axis of difference has the potential to be relevant when studying contextual effects on infant health. The concept of linked fate, which refers to the feeling that one's own life chances are impacted by what happens to other members of one's group, suggests that the mechanism connecting police violence to infant health outcomes may differ between native-born and immigrant Black populations (23). For instance, these populations may differ in the types of events they interpret as salient attacks on their group and consequently experience as sources of stress. Traditionally, the widespread view has been that Black Americans shared a sense of group

consciousness—a linked fate—with other Black Americans based on their racial group's shared historical experiences in the United States (23, 24). Similarly, Black immigrant populations may also share a form of linked fate unique from that of native-born Black Americans (22–24).

Immigrant Black Americans and native-born Black Americans often experience distinct cultural perspectives, historical roots, and socialization regarding race in the United States. Native-born Black Americans' identities are shaped by centuries-old American history, including slavery, segregation, and the ongoing fight for civil rights (23, 25). They often view American society through a lens of deep-seated, systemic domestic racism (26). Black immigrants grew up in different parts of Africa or the Caribbean with distinct ethnic, tribal, and national identities. They often arrive with an "immigrant optimism"—the belief that hard work directly translates to success—often unburdened by early socialization regarding American racial hostility (24, 27).. Many newly arrived Black immigrants initially define themselves by their home country or continent rather than a collective American racial category (23, 24). African and Caribbean immigrants also rely on strong transnational, extended family, and community-based social networks to navigate life in the United States. These material and symbolic differences have implications for stress appraisal and, therefore, for how immigrant Black parents may differ in their responses to police violence.

On the other hand, while Black immigrant populations may be differentiated from native-born Black populations along many dimensions (e.g., education), they are not immune to police racial profiling and violence, especially in fluid, unpredictable, high stakes and chaotic situations (22, 27). While most police victimizations involve native-born Black Americans, cases involving Black immigrants are also prolific (28), and very rarely has research considered the latter. By remaining nativity-neutral with regards to police killings, we thus cannot know whether and how, for example, police violence involving Black immigrants comes to affect foreign-born Black parents and the life chances of their children. More broadly, research that fails to consider this heterogeneity misses an opportunity to produce a finer theory of policing as a stressor.

To further understand the nuances of the link between police violence and infant health, and in an effort to address understudied heterogeneity between immigrant and native-born Black populations, we examine one particular case of police brutality: the killing of Amadou Diallo in the Bronx, New York in 1999. In the early hours of 4 February 1999, four plainclothes New York Police Department (NYPD) Street Crimes Unit on patrol see Amadou Diallo, a 23-year-old Black male immigrant from Guinea, standing outside the vestibule of his Bronx apartment. The officers stop their unmarked vehicle nearby, exit, and approach him. They testify that Diallo began to step back while clasping a "black square object" in his pocket. Mistaking the object for a gun, one officer panics and fires his semiautomatic pistol, and the rest follow suit. In total, the unit fires 41 rounds on Diallo, 19 of which make contact. The officers frisk the body, but there was no gun: Diallo had been reaching for his wallet. News of the fusillade spread quickly in New York, and as public interest grew, that Diallo was a young immigrant became a salient part of his case. Shared widely was the anecdote about how Diallo, the night of his killing, had told his mother Kadiatou he had saved enough money for college (29, 30). For a sympathetic public, there was a compelling narrative that Diallo was an "optimistic immigrant" dispossessed of his ambitions by violence all the same (29, 31, 32). His death was evidence that Black immigrants were no less inhabitants of policing regimes, and that American policing could have fatal and far-reaching consequences for Black foreign- and native-borns alike.

We ask whether, given its physical proximity, the killing of Amadou Diallo had an effect on birth outcomes for non-Hispanic (NH) Black infants in the Bronx, and if, given Diallo's own status as an immigrant, this effect was particularly concentrated within the foreign-born NH Black population. Specifically, we examine the odds of a male birth to foreign-born NH Black mothers in the Bronx. Because male fetuses are more susceptible to miscarriage and premature death than female fetuses (33), sex ratios of live births are an infant health indicator that has been shown to be responsive to a wide range of environmental and contextual factors which increase or decrease maternal stress (33, 34).

Using data on live infants born to residents of the Bronx, New York from the National Vital Statistics System, we analyze more than 50,000 births, aggregated by sex and race/ethnicity, over a test period of 68 months spanning January 1996 to August 2001. Precise details about the timing of the killing, as well as comprehensive information about mothers' place of residence and country of birth, enable us to study the effect of Diallo's death on male births to the foreign-born NH Black population in the Bronx. We employ an interrupted time-series approach, hypothesizing that the monthly odds of a male birth to a foreign-born NH Black mother would fall below its expected value in birth cohorts exposed *in utero* to the Amadou Diallo shooting.

ANALYTICAL APPROACH

Data and Variables

We retrieved information on live infants born to residents of the Bronx Borough, New York City, from the National Vital Statistics System (NVSS). The NVSS contains all live births registered by state vital statistics agencies, who then report these data to the federal agency. NVSS makes restricted-used live birth data (i.e., with identifiers such as residential county of birth) available to researchers under a data use agreement. We acquired NVSS data on live births starting with 1996, the year in which most states adopted and transmitted electronic birth certificate data to NVSS. Previous reports describe in detail the quality and provenance of NVSS and the analytic variables used for our study.

The NVSS contains the mother's place of residence, calendar month and year of birth, infant sex, and race/ethnicity of mother as well as country of birth. Using these variables, we aggregated live birth data, by sex and race/ethnicity, to 68 months spanning January 1996 to August 2001. We end the time series in August 2001 because much literature shows perturbations in birth outcomes, including in the odds of a male live birth, following the terrorist attacks of September 11, 2001 (35–37). We therefore retained a sufficiently long time series that permits adequate statistical power to detect potential responses to the Amadou Diallo shooting, while avoiding the potential violation of heteroskedasticity in the series arising from 9/11-related perturbations.

We restricted the geographic catchment area to newborns whose mother holds a residential address in the Bronx Borough. This region housed ~1.2 million residents in 1999 (about 17% of the population of New York City) (38).

We specified as the dependent variable the odds of a male birth to a foreign-born NH Black mother in a calendar month. This variable is similar to a conventional sex ratio measure (i.e., odds of a male birth, which is identical to the ratio of males to females) but incorporates into the denominator live births in "control" populations who, according to our hypothesis, would remain unaffected (or less affected) by the Amadou Diallo shooting. These control

populations include all births to NH White and US-born NH Black residents of the Bronx, as well as foreign-born NH Black females. The benefit of including these control populations (when calculating the odds) involves control for confounding by any unmeasured "third variable" that increases (or decreases) overall live births in the Bronx for any particular month (or months). We also take the natural logarithm of the odds metric to (i.) limit the threat of non-constant variance and (ii.) render a more straightforward interpretation of the coefficients of the independent variables (which, when exponentiated, produce an "effect on odds" metric familiar to epidemiologists and others).

Methods

Our approach follows the tradition of the interrupted time-series quasi-experiment (39, 40). We hypothesize that the monthly odds of a NH Black male birth to a foreign-born NH Black mother falls below its expected value in cohorts exposed *in utero* to the Amadou Diallo shooting. The odds of a male birth, however, exhibits well-documented temporal patterns (e.g., seasonality, trend, and the tendency for high or low values to be "remembered" into subsequent months) (41). These patterns, referred to collectively as autocorrelation, confound tests of association because the expected value of a patterned series is not its mean.

To address the autocorrelation problem, researchers have devised data-driven routines that identify and remove patterns in the dependent variable. We used the routines devised by Box and Jenkins (42) to implement this approach. These routines express autocorrelation as a product of "autoregressive" (AR), "integrated" (I) and "moving average" (MA) parameters, collectively referred to as ARIMA models. The residuals of these ARIMA models meet the assumptions of correlational tests in that the expected value is zero and the monthly observations are statistically independent of one another. This allows us to isolate and analyze any variation in odds of a male birth to a foreign-born NH Black mother that is *not* part of the expected temporal pattern.

We proceeded through six steps. First, we estimated initial models for the entire series (68 months beginning January 1996) using software from Scientific Computing Associates (version 5.4.6, SCA Corp., Villa Park, IL), in which we specified a constant as the only "right side" variable. Second, we inspected the error term for autocorrelation and, if needed, inserted Box-Jenkins ARIMA parameters to remove any temporal patterns in the log-odds of a male live birth to a foreign-born NH Black mother. Third, we added the binary independent variable (i.e., Amadou Diallo shooting, coded as 1 in February 1999 and 0 otherwise) to the best-fitting Box-Jenkins models of the log-odds of a male birth to a foreign-born NH Black mother. We inspected a concurrent response as well as lags of up to four months (i.e., Feb 1999 through to June 1999) to ensure capturing any delayed associations. This time lag coheres with literature reporting male-specific responses to population shocks within a few months (43). Fourth, we inspected the residual values of the error term to ensure that they exhibited no temporal patterns. Fifth, we performed sensitivity tests and assessed the stability of results to outliers (44) as well as to the examination of lagged associations through to Oct 1999 (i.e., eight months later). Sixth, we repeated all steps for one additional dependent variable: the odds of a male birth to a native-born NH Black mother. Consistent with the notion that male fetuses among foreign-born NH Black mothers appear particularly sensitive to the Amadou Diallo shooting, we hypothesize no change in the odds of a male birth to a native-born NH Black mother following the shooting.

RESULTS

Table 1 provides descriptive statistics of 53,714 births to both foreign- and native-born NH white and NH Black mothers in the Bronx Borough over the test period. Males account for 51 percent of live births. The mean monthly count of male births among foreign-born NH Black mothers is 130.6. This count shows a wide range across the test period (i.e., 97 to 164 live births).

[Table 1 here]

Inspection of the autocorrelation and partial autocorrelation functions (ACF and PACF) of the dependent variable revealed that high values were "remembered" 1 and 3 months later, albeit in diminishing amounts. We inserted an autoregressive (AR) parameter at lag 1 month and a moving average (MA) parameter at lag 3 months to parsimoniously control for this autocorrelation. Figure 1 plots the log-odds of a male birth to a foreign-born NH Black mother for the 68 months beginning January 1996. Panel A shows the observed log-odds (with mean of -1.62, which equates to an odds of 0.198). Panel B illustrates the residual log-odds after removal of autocorrelation (the first month of Panel B was lost to ARIMA modeling). Visual inspection suggests that the residual log-odds appears low in February 1999 and March 1999. This indicates a possible decrease in the odds of a male birth to a foreign-born NH Black mother in the months following the shooting, even after adjusting for the temporal patterning of births; we formally test this possibility below.

[Figure 1 about here]

Table 2 shows the results from our final test equation for the log-odds of a male birth to a foreign-born NH Black mother. After including the AR(1) and MA(3) parameters, none of the lags of the residual values of the dependent variable exceeded twice their standard error (Supplemental Material, Table i.). We, therefore, required no additional ARIMA parameters.

[Table 2 about here]

Male fetuses among foreign-born NH Black mothers in February 1999 and March 1999 show a lower than expected odds of a live birth (log-odds coefficient for Feb 1999 = -.177, standard error [SE]: .085; coef. for Mar 1999 = -.202, SE: .090, $p<.05$ for each). We cannot reject the null for any other lag following the Feb 1999 shooting.

To give the reader a sense of the magnitude of the discovered relation, we estimated (from Table 2) the number of male births among foreign-born NH Black mothers statistically "lost" to the Feb 1999 Amadou Diallo shooting. Taking the antilog of the Feb 1999 and Mar 1999 coefficients in Table 2 produces odds ratios of 0.84 and 0.82, respectively. We then applied these odds ratios to the count of live male births (the numerator) from the mean odds of male births among foreign-born NH Black mothers in the test period. This calculation shows 45 male births statistically "lost" to the Amadou Diallo shooting. When applied to the "expected" count of males born in these two months, the Amadou Diallo shooting statistically accounts for a 17% drop in the number of male births among foreign-born NH Black mothers.

We conducted two additional tests to gauge the robustness of our findings. First, we applied routines (*44*) that detect and control outliers in the dependent variable to assess whether they reduced the efficiency of our estimates by inflating confidence intervals. These routines detected no outliers. Second, as a sensitivity check, we examined whether a male-specific response occurred among male births among native-born NH Black mothers as well. Given that the logic of our Introduction presumes a decline in males specifically among births among foreign-born NH Black mothers, we expected no relation (or an attenuated relation) between the shooting and the odds of a male birth to a native-born NH Black mother. Time-series results for this group (Table 3), consistent with this logic, show no relation.

[Table 3 about here]

DISCUSSION

Amadou Diallo was killed on February 4, 1999, fourteen years before the Black Lives Matter movement began in 2013. While it did not happen during the Black Lives Matter era, Diallo's tragic death is widely considered a foundational case. It brought broad attention to police brutality and racial profiling (29–32), laying the groundwork for future research on its social, political, economic, and health consequences.

The death of Amadou Diallo was not only extensively shared in the media but reverberated through West African and wider Black immigrant networks in New York City, particularly the Bronx. His death suggested that Black immigrant populations were not immune to police violence. Many West African immigrants previously viewed themselves as separate from domestic African American struggles. Diallo's death provided evidence that police racial profiling targeted all Black bodies regardless of immigration status or nationality, making his death particularly impactful to the Black foreign-born community. Our results largely support this view as it pertains to population health: we find a significant reduction in the odds of a male birth to a foreign-born NH Black mother in the two months following the killing of Amadou Diallo.

However, the implications of our work must be considered. While Novak et al. (2017) report increases in risks of low birthweight among infants born to both foreign-born and native-born Latina mothers following a highly public federal immigration raid, a racialized stressor analogous to the Diallo killing (45), our lack of a detectable effect on male births to native-born NH Black mothers warrants further investigation. Since we are unable to capture father-specific information in the NVSS, such as the father’s race/ethnicity and place of birth, future work might consider the role of household composition, and whether, for example, mixed-nativity Black households obtain different responses to police violence compared to same-status households. More broadly, our work demonstrates the greater effort needed in incorporating different kinds of public police violence as analytical cases, and in describing variation between foreign-born and native-born co-ethnics that unfolds as a result. For example, because police violence varies case-to-case, questions persist over whether it is perhaps the epistemic quality that any police violence happened at all that makes it an ecological stressor for foreign-born versus native-born Black parents, or if there are event-specific features of police violence that render some incidents stressful and others not. Future work, leveraging multiple sources of policing as variables, should therefore test and refine these conceptual boundaries.

Despite these limitations, our findings expand the literature by suggesting the need to consider heterogeneity within the Black population when examining the drivers of population health and health disparities. Black is not a monolith. Differential histories of incorporation into the United States have implications for stress appraisal, as they do for other social and demographic processes (46, 47). Future work that analyzes the impact of highly publicized police killings should account for this heterogeneity, as well as attend to its historical context, such as the advent and the preponderance of social media platforms in recent years for sharing these events.

TABLES AND FIGURES

Table 1. Characteristics of 53,714 NH Black and NH White Live Births in the Bronx, NY over 68 Months Spanning January 1996 to August 2001.

	n	Mean Monthly Count	Range (Min – Max)
Male	27,394	—	—
NH Black, Foreign-Born	8,882	130.6	97 – 164
NH Black, US-Born	12,859	189.1	147 – 232
NH White	5,653	83.1	61 – 120
Female	26,320	—	—
NH Black, Foreign-Born	8,611	126.6	100 – 158
NH Black, US-Born	12,444	183.0	144 – 214
NH White	5,265	77.4	54 – 111

Table 2. Time series results predicting the log-odds of a male birth to a *foreign-born* non-Hispanic Black mother in the Bronx, NY, from January 1996 to August 2001 as a function of autocorrelation and the Amadou Diallo shooting (n=68; standard errors [SE] in parentheses). Coefficients for the Amadou Diallo shooting represent differences in log-odds.

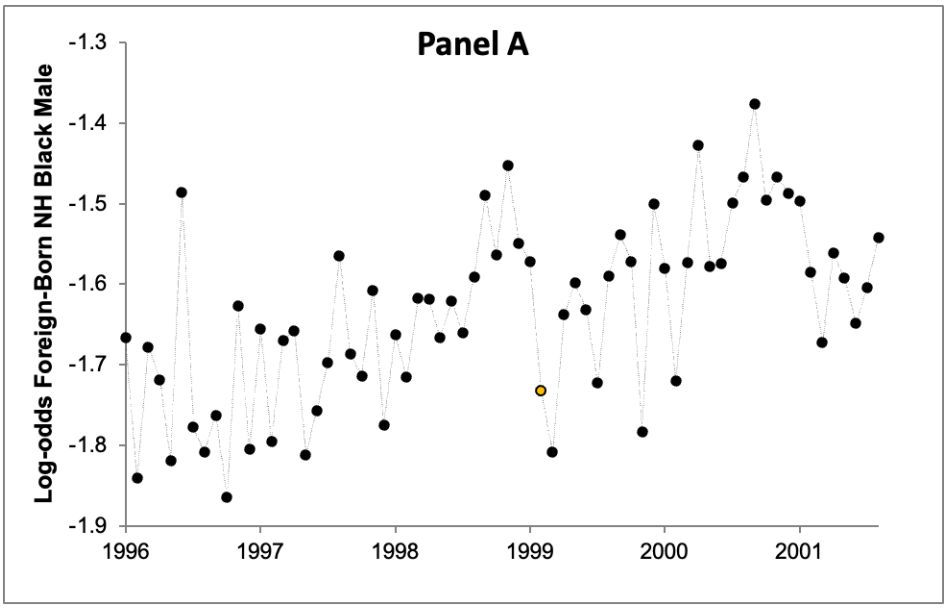
Parameter	**Lag (months)**	**Predicted log-odds of a male birth to a foreign-born NH Black mother**	
		Coef. (SE)	(Trimmed)
Constant	—	-1.62 (.024)	-1.63 (.023)
Autoregressive Parameter	1	-.378 (.124)	.296 (.115)
Moving Average Parameter	3	-.340 (.130)	-.472 (.113)
Amadou Diallo shooting (Feb 1999)	0	-.177 (.085)*	-.176 (.080)*
	1	-.202 (.090)*	-.185 (.080)*
	2	-.029 (.091)	
	3	.017 (.098)	
	4	-.009 (.099)	
	5	-.087 (.098)	
	6	.077 (.091)	
	7	.064 (.090)	
	8	.101 (.084)	

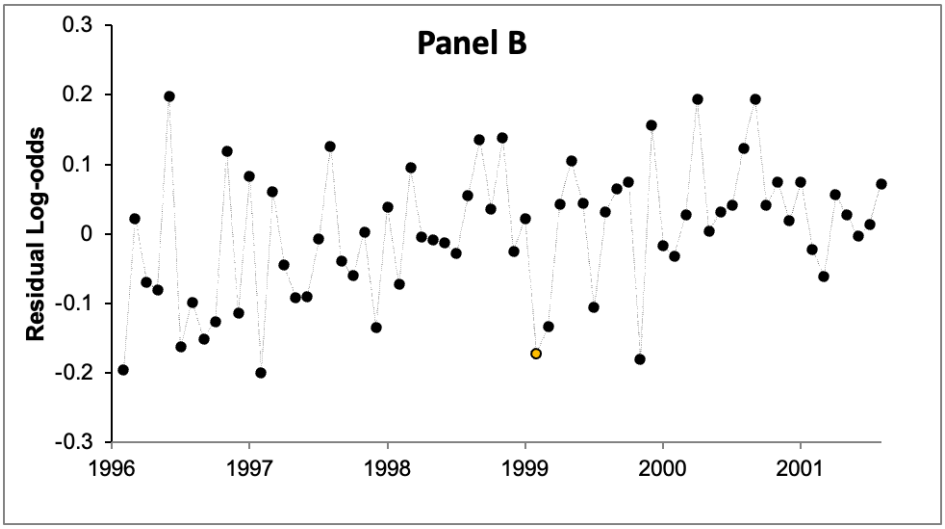
*$p < 0.5$

Table 3. Sensitivity Check. Time series results predicting the log-odds of a male birth to a *US-born* non-Hispanic Black mother in the Bronx, NY, from January 1996 to August 2001 as a function of autocorrelation and the Amadou Diallo shooting (n=68; standard errors [SE] in parentheses). Coefficients for the Amadou Diallo shooting represent differences in log-odds.

Parameter	**Lag (months)**	**Predicted log-odds of a male birth to a US-born NH Black mother**
		Coef. (SE)
Constant	—	-1.18 (.018)
Autoregressive Parameter	3	.388 (.124)
Amadou Diallo shooting (Feb 1999)	0	.147 (.077)
	1	.101 (.090)
	2	-.065 (.091)
	3	-.022 (.098)
	4	-.118 (.099)
	5	-.007 (.098)
	6	.127 (.091)
	7	-.015 (.090)
	8	-.003 (.084)

Figure 1. Log-odds of a foreign-born non-Hispanic Black male birth over 68 months in the Bronx Borough, New York City, Jan 1996 to Aug 2001. Panel A plots the observed log-odds; Panel B plots the residual log-odds, with mean=0, after removal of autocorrelation (first 3 months lost to time-series modeling). February 1999 is indicated with an orange circle.





REFERENCES

1. A. P. Thakrar, A. D. Forrest, M. G. Maltenfort, C. B. Forrest, Child Mortality In The US And 19 OECD Comparator Nations: A 50-Year Time-Trend Analysis. *Health Aff. (Millwood)* **37**, 140–149 (2018).
2. D. M. Ely, A. K. Driscoll, "Infant Mortality in the United States, 2023: Data From the Period Linked Birth/Infant Death File" (Volume 74, Number 7, National Vital Statistics System, 2025); https://www.cdc.gov/nchs/data/nvsr/nvsr73/nvsr73-05.pdf.
3. J. F. Culhane, R. L. Goldenberg, Racial Disparities in Preterm Birth. *Semin. Perinatol.* **35**, 234–239 (2011).
4. J. Crear-Perry, R. Correa-de-Araujo, T. Lewis Johnson, M. R. McLemore, E. Neilson, M. Wallace, Social and Structural Determinants of Health Inequities in Maternal Health. *J. Womens Health* **30**, 230–235 (2021).
5. H. Lee, How Does Structural Racism Operate (in) the Contemporary US Criminal Justice System? *Annu. Rev. Criminol.* **7**, 233–255 (2024).
6. K. G. Muhammad, *The Condemnation of Blackness: Race, Crime, and the Making of Modern Urban America, With a New Preface* (Harvard University Press, Cambridge, MA, 2019).
7. M. K. Chen, K. L. Christensen, E. John, E. Owens, Y. Zhuo, Smartphone Data Reveal Neighborhood-Level Racial Disparities in Police Presence. *Rev. Econ. Stat.* **107**, 1734–1742 (2025).
8. F. Edwards, H. Lee, M. Esposito, Risk of being killed by police use of force in the United States by age, race–ethnicity, and sex. *Proc. Natl. Acad. Sci.* **116**, 16793–16798 (2019).
9. R. Mehra, A. Alspaugh, L. S. Franck, M. R. McLemore, T. S. Kershaw, J. R. Ickovics, D. E. Keene, A. A. Sewell, "Police shootings, now that seems to be the main issue" – Black pregnant women's anticipation of police brutality towards their children. *BMC Public Health* **22**, 146 (2022).
10. S. Alang, D. McAlpine, E. McCreedy, R. Hardeman, Police Brutality and Black Health: Setting the Agenda for Public Health Scholars. *Am. J. Public Health* **107**, 662–665 (2017).
11. A. A. Sewell, J. M. Feldman, R. Ray, K. L. Gilbert, K. A. Jefferson, H. Lee, Illness spillovers of lethal police violence: the significance of gendered marginalization. *Ethn. Racial Stud.* **44**, 1089–1114 (2021).
12. J. E. DeVylder, D. M. Anglin, L. Bowleg, L. Fedina, B. G. Link, Police Violence and Public Health. *Annu. Rev. Clin. Psychol.* **18**, 527–552 (2022).
13. A. A. Sewell, The Illness Associations of Police Violence: Differential Relationships by Ethnoracial Composition. *Sociol. Forum* **32**, 975–997 (2017).
14. R. Haile, T. Rowell-Cunsolo, M.-F. Hyacinthe, S. Alang, "We (still) charge genocide": A systematic review and synthesis of the direct and indirect health consequences of police violence in the United States. *Soc. Sci. Med.* **322**, 115784 (2023).
15. E. Rock, L. Beletsky, B. del Pozo, Policing as a Structural Determinant of Health. *Annu. Rev. Public Health* **47**, 235–247 (2026).
16. N. Dole, D. A. Savitz, I. Hertz-Picciotto, A. M. Siega-Riz, M. J. McMahon, P. Buekens, Maternal stress and preterm birth. *Am. J. Epidemiol.* **157**, 14–24 (2003).
17. T. A. Manuck, Racial and ethnic differences in preterm birth: A complex, multifactorial problem. *Semin. Perinatol.* **41**, 511–518 (2017).

18. F. Torche, The effect of maternal stress on birth outcomes: exploiting a natural experiment. *Demography* **48**, 1473–1491 (2011).
19. R. Hardeman, T. Chantarat, J. Karbeah, Police Exposure As a Determinant of Structural Racism: An Exploration of the Association between Preterm Birth in Neighborhoods with High Police Exposure. *Health Serv. Res.* **55**, 50–50 (2020).
20. D. E. Goin, A. M. Gomez, K. Farkas, C. Duarte, D. Karasek, B. D. Chambers, A. V. Jackson, J. Ahern, Occurrence of fatal police violence during pregnancy and hazard of preterm birth in California. *Paediatr. Perinat. Epidemiol.* **35**, 469–478 (2021).
21. J. L. Jahn, N. Krieger, M. Agénor, M. Leung, B. A. Davis, M. G. Weisskopf, J. T. Chen, Gestational exposure to fatal police violence and pregnancy loss in US core based statistical areas, 2013-2015. *EClinicalMedicine* **36**, 100901 (2021).
22. C. W. Smith, *Black Mosaic: The Politics of Black Pan-Ethnic Diversity* (NYU Press, 2014).
23. S. C. Nunnally, LINKING BLACKNESS OR ETHNIC OTHERING?: African Americans' Diasporic Linked Fate with West Indian and African Peoples in the United States. *Bois Rev. Soc. Sci. Res. Race* **7**, 335–355 (2010).
24. M. C. Waters, Ethnic and Racial Identities of Second-Generation Black Immigrants in New York City. *Int. Migr. Rev.* **28**, 795–820 (1994).
25. M. C. Dawson, *Behind the Mule: Race and Class in African-American Politics* (Princeton University Press, 1994).
26. M. T. Price, *Dreaming Blackness: Black Nationalism and African American Public Opinion* (NYU Press, 2009; https://www.jstor.org/stable/j.ctt9qft04).
27. C. M. Greer, *Black Ethnics: Race, Immigration, and the Pursuit of the American Dream* (Oxford University Press, Oxford, New York, 2013).
28. N. Foner, Ed., *New Immigrants in New York: Completely Revised and Updated Edition* (Columbia University Press, 2001).
29. C. Haberman, 41 Shots, Then 6 Years of Healing, *The New York Times* (2005). https://www.nytimes.com/2005/02/04/nyregion/41-shots-then-6-years-of-healing.html.
30. C. Red, Years Before Black Lives Matter, 41 Shots Killed Him, *The New York Times* (2019). https://www.nytimes.com/2019/07/19/nyregion/amadou-diallo-mother-eric-garner.html.
31. M. Tran, NEIGHBORHOOD REPORT: SOUNDVIEW; Where Diallo Lived, Soothing and Bearing Witness, *The New York Times* (2004). https://www.nytimes.com/2004/02/01/nyregion/neighborhood-report-soundview-where-diallo-lived-soothing-and-bearing-witness.html.
32. A. Waldman, A Hard Worker With a Gentle Smile, *The New York Times* (1999). https://www.nytimes.com/1999/02/05/nyregion/a-hard-worker-with-a-gentle-smile.html.
33. M. M. McCarthy, Stress during pregnancy: Fetal males pay the price. *Proc. Natl. Acad. Sci.* **116**, 23877–23879 (2019).
34. R. Schacht, D. Tharp, K. R. Smith, Sex ratios at birth vary with environmental harshness but not maternal condition. *Sci. Rep.* **9**, 9066 (2019).
35. B. Eskenazi, A. R. Marks, R. Catalano, T. Bruckner, P. G. Toniolo, Low birthweight in New York City and upstate New York following the events of September 11th. *Hum. Reprod.* **22**, 3013–3020 (2007).
36. R. Catalano, T. Bruckner, A. R. Marks, B. Eskenazi, Exogenous shocks to the human sex ratio: the case of September 11, 2001 in New York City. *Hum. Reprod.* **21**, 3127–3131 (2006).

37. T. A. Bruckner, R. Catalano, J. Ahern, Male fetal loss in the U.S. following the terrorist attacks of September 11, 2001. *BMC Public Health* **10**, 273 (2010).
38. Table 2: Estimated 1999 Population, 1990 Census, 1980 Census, Land Area and Population Density by Resident County, New York State, *Vital Statistics of New York State 1999 Tables* (2001). https://www.health.ny.gov/statistics/vital_statistics/1999/table02.htm.
39. J. L. Bernal, S. Cummins, A. Gasparrini, Interrupted time series regression for the evaluation of public health interventions: a tutorial. *Int. J. Epidemiol.* **46**, 348–355 (2017).
40. U. Helfenstein, The use of transfer function models, intervention analysis and related time series methods in epidemiology. *Int. J. Epidemiol.* **20**, 808–815 (1991).
41. T. A. Bruckner, É. Lebreton, N. Perrone, L. H. Mortensen, B. Blondel, Preterm birth and selection in utero among males following the November 2015 Paris attacks. *Int. J. Epidemiol.* **48**, 1614–1622 (2019).
42. G. E. P. Box, G. M. Jenkins, G. C. Reinsel, G. M. Ljung, *Time Series Analysis: Forecasting and Control* (John Wiley & Sons, 5th Edition., 2015).
43. T. A. Bruckner, R. Catalano, Selection in utero and population health: Theory and typology of research. *SSM - Popul. Health* **5**, 101–113 (2018).
44. I. Chang, G. C. Tiao, C. Chen, Estimation of Time Series Parameters in the Presence of Outliers. *Technometrics* **30**, 193–204 (1988).
45. N. L. Novak, A. T. Geronimus, A. M. Martinez-Cardoso, Change in birth outcomes among infants born to Latina mothers after a major immigration raid. *Int. J. Epidemiol.* **46**, 839–849 (2017).
46. A. Portes, R. G. Rumbaut, *Immigrant America: A Portrait* (University of California Press, ed. 4, 2014).
47. D. S. Massey, *Categorically Unequal: The American Stratification System* (Russell Sage Foundation, 2007).